Zilong Zhang

Ph.D. Student in Statistics and Machine Learning, Georgia State University
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EDUCATION

Ph.D. in Machine Learning & Data Science **Aug. 2025 – Present**
Department of Mathematics & Statistics / Georgia State University *Atlanta, GA*

M.S. in Statistics **Aug. 2023 – May 2025**
Department of Statistics & Probability / Michigan State University *East Lansing, MI*

Bachelor of Financial Mathematics **Sep. 2019 – Jul. 2023**
School of Statistics / University of International Business and Economics *Beijing, China*

PROFILE

- Statistics and machine-learning Ph.D. researcher focused on extracting reliable signals from noisy, high-dimensional data and evaluating model decisions when verification is limited.
- Interested in product and advertising measurement, decision quality, trustworthy AI, selective-label learning, and functional data analysis; brings statistical rigor, Python/R analysis, and experience communicating findings through research and financial due-diligence work.

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Weiyi He, Junxi Zhang, Lei Ding, Chi-Kuang Yeh. "AURA: Adaptive Uncertainty-aware Refinement for LLM-as-a-Judge Auditing" *Under review, arXiv:2606.19714*, 2026.
- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. "Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning" *Under review, arXiv:2606.19057*, **2026**.

RESEARCH AND OTHER PROJECTS

Uncertainty-Aware Auditing of LLM-as-a-Judge Systems **2026**

- Developed an approach for using limited human comparisons efficiently by estimating judgment reliability, carrying forward trusted pairwise evidence, and surfacing uncertain cases for review; manuscript under review.

Selective-Label Learning for LLM Evaluation **2026**

- Formulated selectively human-verified pairwise preferences as a positive–unlabeled learning problem; explored partial optimal transport to match verified positives with reliable unlabeled comparisons and reduce judge/verbosity bias without retraining the judge.

Functional Time-Series Research **Sept. 2025 – Present**

- Research on bandwidth selection for functional time series and long-run covariance estimation.

mRNA-to-Protein Prediction with Transformers and GNNs **Jan. 2024 – Jan. 2026**

- Developed a Transformer/GNN approach for predicting protein expression from single-cell RNA measurements, working with collaborators at Michigan State and MD Anderson on cross-institution biomedical ML.

WORK EXPERIENCE

China Securities Co., Ltd. **Jul. 2021 – Nov. 2021**
Investment Management Intern *Beijing, China*

- Prepared industry research, risk assessments, listing materials, and fundraising reports for client companies.

TECHNICAL SKILLS

Programming and data: Python, R, C, scikit-learn, NumPy, pandas
Machine learning: PyTorch, TensorFlow, Hugging Face Transformers, GNNs
Methods and tools: statistical learning, data analysis, LLM evaluation; Git, Linux, Jupyter, VS Code